A 45-Year Global Analysis of the Spatial Human Forest Nexus

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Abstract

Forests play a crucial role in providing important ecosystem services such as biodiversity conservation, carbon sequestration, and biomass production, while supporting the livelihoods of millions of people globally. Understanding the spatial relationship between human populations and forests is vital for assessing how these socio-environmental systems interact and evolve over time. However, a global scale understanding of these spatial relationships remains limited. Here, we address this gap by examining the joint global trends in human population and forest dynamics from 1975 to 2020. We assess two indicators of spatial and temporal trends: Forest Area per Person (FAP) and Forest Proximate People (FPP), illustrating how the spatial interplay between humans and forests has changed over time. We then introduce the Forest Human Nexus (FHN), a spatial indicator that integrates forest area per capita, forest accessibility, and population density to provide a geospatial assessment of human-forest relationships. Our analysis reveals an uneven distribution of FHN trends worldwide, reflecting severe ecological stress in many tropical regions and highlighting the need for targeted interventions. By offering detailed spatiotemporal insights and a new spatial metric for human-forest interactions, this study aims to inform policymakers, conservationists, development practitioners, and researchers in fostering more sustainable human-forest relationships.

Introduction

Forests cover 31% of the of the land area on our planet[1] and play a crucial role as complex socio-environmental systems, providing essential ecosystem services such as carbon storage[2] and sequestration[3], and biodiversity conservation[4], while also supporting the livelihoods of millions of people globally[5]. The spatial dimension of human-forest interactions has evolved significantly over recent decades due to global land use changes, urbanization, and demographic shifts. Studies have documented substantial changes in the spatial distribution of forest-adjacent communities, with some regions experiencing migration away from forests while others see increasing settlement near forest edges[6]. For example, research in the Brazilian Amazon has shown complex patterns of colonization along forest frontiers, followed by urbanization processes that reshape human-forest spatial relationships[7]. In Southeast Asia, the expansion of plantation agriculture has transformed traditional forest-adjacent communities, altering centuries-old spatial configurations of human-forest coexistence[8].

Indeed, around one-fourth of the global population (over 1.6 billion people) live within 5km of a forest[9]. In some areas, such as parts of Europe and North America, the number of people living near forests has actually increased as suburban development expands into forested landscapes[10]. Conversely, in regions experiencing rapid deforestation, forest-adjacent populations may either migrate away from degraded forest margins or remain in place while the forests recede, fundamentally altering their spatial relationship with forest ecosystems[11]. Understanding these dynamic spatial patterns is crucial for developing effective conservation and development policies that account for the changing human-forest nexus[12]. These landscapes contribute significantly to environmental stability and socio-economic development, offering benefits ranging from health improvements to income generation for those living in or near forested areas[13]. Consequently, the conservation, governance, and restoration of forests, alongside the sustainable development of forest-adjacent communities, are highly important on the global agenda.

Understanding the spatial relationship between human populations and forests is critical for several reasons[14]. Access to forest ecosystem services can enhance people's quality of life[15], support livelihoods[5], and contribute to sustainable development[16]. However, the dynamics of human-forest interactions are complex and multifaceted[17]. On one hand, proximity to forests may enhance ecological awareness[18] and foster conservation initiatives[19]. On the other hand, increased human encroachment into forested areas can lead to habitat degradation, biodiversity loss[20], and forest fragmentation[21], undermining the services forests provide[22]. Additionally, exposure to wild areas brings potential health risks, as interactions with wildlife can facilitate the transmission of zoonotic diseases[23], as evidenced by recent outbreaks[24–26]. Therefore, a comprehensive understanding of how human populations spatially relate to forests is essential, not only to maximise the ecological and socio-economic benefits but also to address the associated health and environmental challenges.

Despite the importance of human-forest relationships, there remain significant gaps in our understanding, particularly on a global scale. Previous studies have often focused on localized[27] or short-term period evidence[9], lacking comprehensive quantitative analyses that encompass large temporal and spatial scales. This paper addresses this gap by examining global trends in human-forest proximity from 1975 to 2020. To do so we used the HILDA+ dataset for land use classification and forest areas[28]. This dataset reports changes in forest area from 1960 to 2019, including dynamics of deforestation and reforestation. We overlaid this dataset with population density from the Global Human Settlement Layer dataset [29] (see Methods section) at the resolution of 1km, and computed multiple metrics in a moving window of a radius of 50km (see Methods section). First, we computed the Forest Area per Person (FAP) metric, which measures the forest area available per person, and the number of Forest Proximate People (FPP)[9], which counts the number of people living within a given distance to the nearest forest. These two metrics and their temporal trends across different regions and periods, quantify how population growth and deforestation have influenced the amount of forest accessible to individuals and communities worldwide. To further evaluate the spatial dimension of human-forest interactions, we introduce the Forest Human Nexus (FHN) index, a composite metric that quantifies the geospatial relationship between human populations and forests. The FHN index is derived as the inverse of the weighted distance from humans to the nearest forest (see Methods section), providing a spatially explicit measure of how human proximity to forests has evolved over time.

In summary, this study provides a spatiotemporal, global-scale analysis of changes in the human-forest nexus over a 45-year period. By quantifying these changes through geospatial metrics, we offer critical insights for policymakers, conservationists, development practitioners, and researchers aiming to understand and address the evolving spatial patterns of human-forest interactions. This work underscores the necessity for targeted spatial planning and interventions to balance human needs with forest conservation, ensuring the sustainability of these landscapes for future generations.

Results

We present a spatio-temporal analysis of three metrics—FAP, FPP, and FHN—for 2020, as well as the relative differences and trends between 1975 and 2020 at pixel level (50km × 50km resolution) and aggregated for macro-regions across continents. More detailed temporal analysis conducted at 5-year intervals between 1975 and 2020 (presented in Supporting Information) revealed no significant inflection points or trend changes for most macro-regions examined. This confirms our focus on the endpoints of this 45-year period, while highlighting that the significant differences in trends and patterns occur predominantly between macro-regions rather than within the temporal evolution of individual regions.

Global analysis of Forest Area per Person

In 2020, the average forest area per person globally was approximately 1 hectare (Fig. 1), which represents a substantial reduction since 1975. The period from 1975 to 2020 saw a marked decrease in global Forest Area per Person (FAP) (Fig. 2). This decline reflects the combined impacts of human population growth and deforestation. We found clear trends in the relationship between forest area and human population at both the global and regional levels (Fig. 2).

Our temporal analysis revealed heterogenous relative changes in FAP across regions (Fig. 2 and SI). Middle Africa experienced the most severe decline, with a 77.8% reduction in FAP between 1975 and 2020, followed closely by Western Africa (-74.0%) and Eastern Africa (-73.6%). These striking relative changes strongly correlate with the Sen's slope analysis, which identified these same regions as having the steepest negative slopes (Middle Africa: -8.38, Eastern Africa: -7.97, Western Africa: -7.78, all $p < 0.001$), confirming the robustness of the observed trends through multiple analytical approaches (SI).

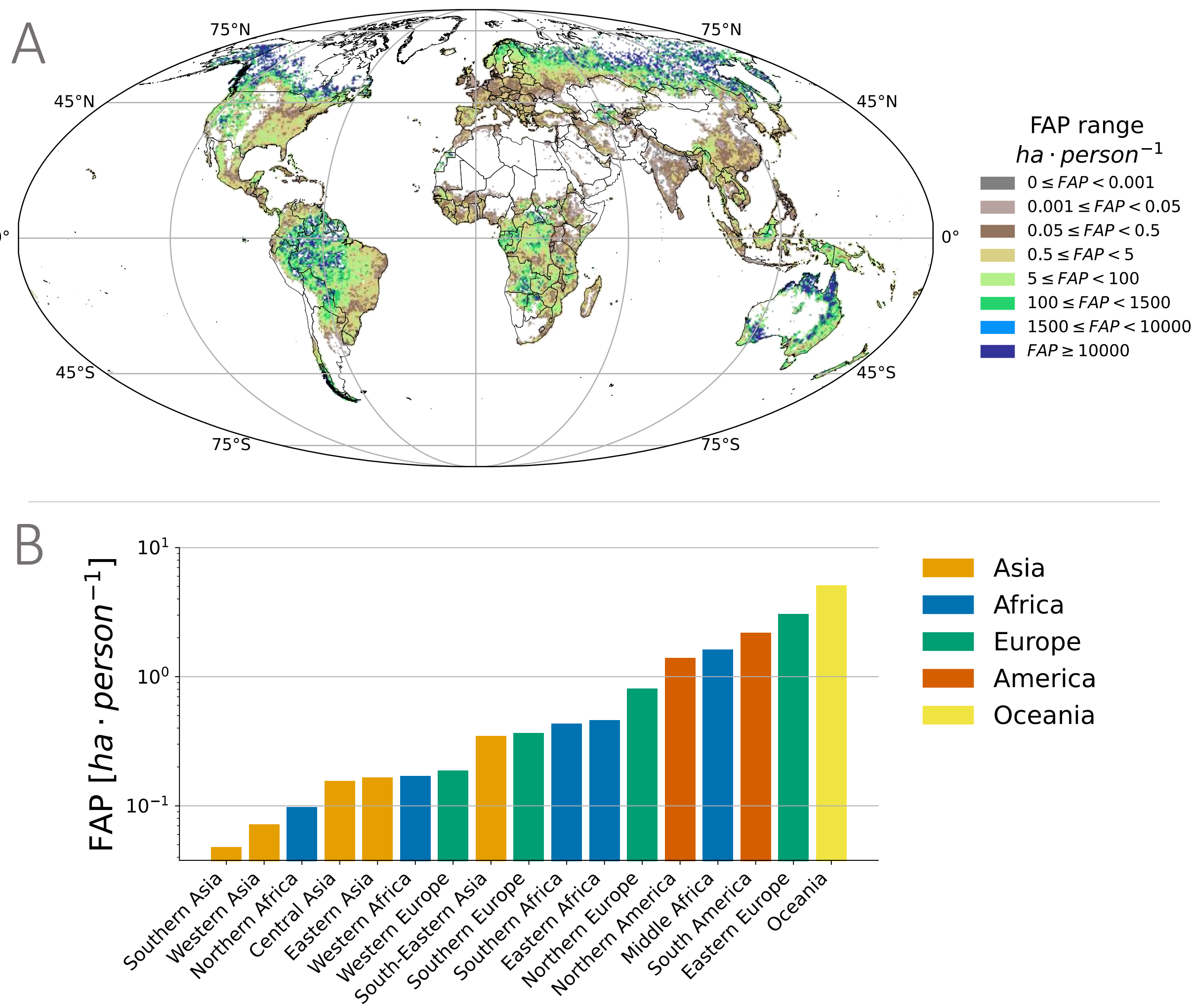

Figure 1. Global distribution of forest area per person (FAP) in 2020. A) Forest Area per Capita observed in 2020 globally and B) for macroregions.

Southern Europe stands out as the only world region that showed a consistent increase of FAP (24.5% relative increase from 1975 to 2020), with Western Europe showing minimal change (+1.0%). This pattern is further confirmed in our Sen's slope analysis (Table S1), which identified Southern Europe as the only region with a significant positive trend (slope: 2.68, $p < 0.01$). These European patterns are due to stable or decreasing populations combined with increasing forest area (Fig. S3 and Fig. S4).

Some of the areas with the highest absolute values of FAP in 2020 (e.g., South America and Middle Africa) were also among those that experienced the greatest declines in FAP, due to deforestation and population growth (Fig. 2A).

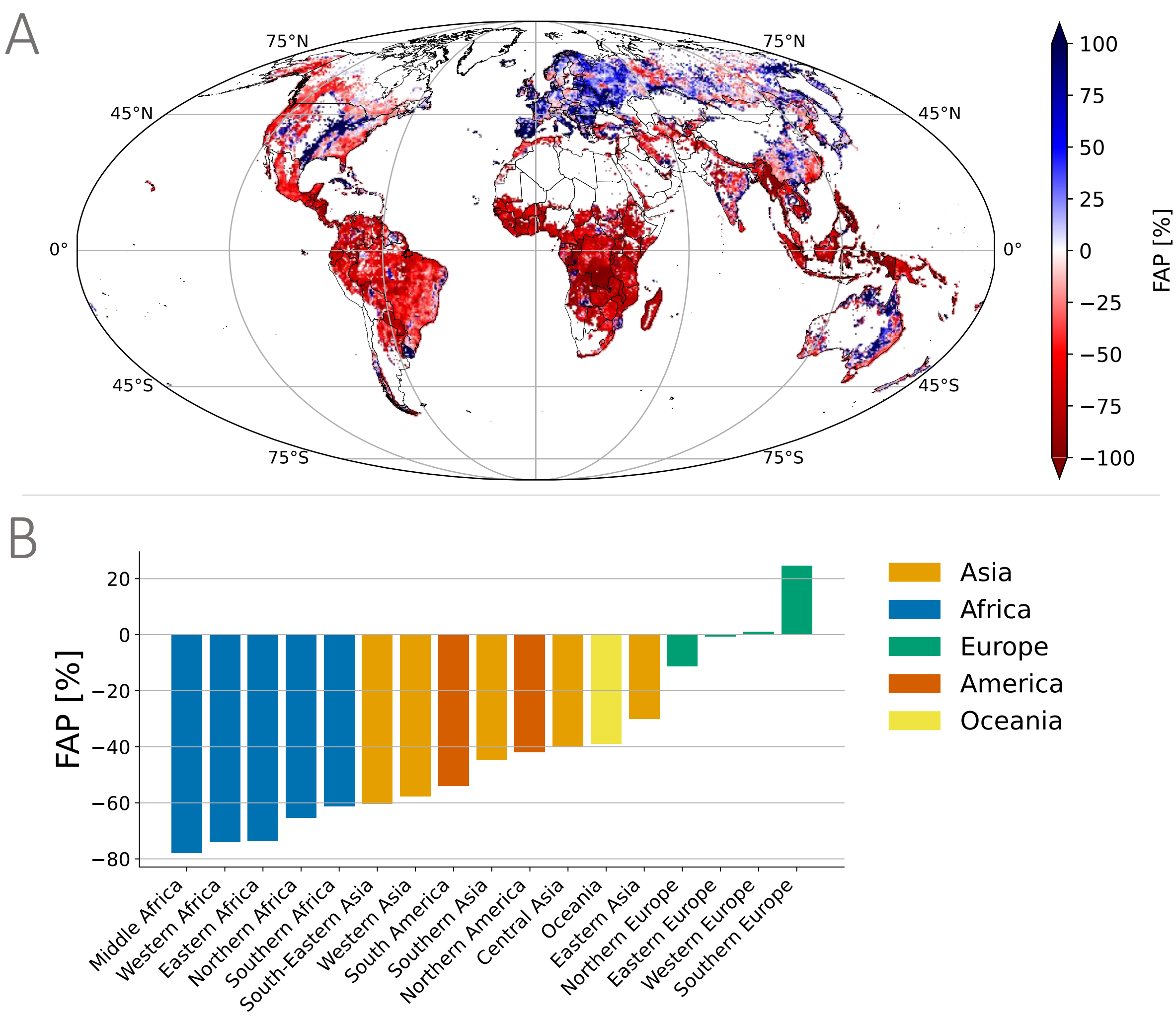

Figure 2. Global change of forest area per person (FAP) between 1975 and 2020. A) Relative change of FAP trend between 2020 and 1975 globally. B) Relative change of median values of FAP across macroregions between 2020 and 1975.

Despite these large declines, these regions still maintained relatively high FAP values in 2020 because of their large initial forest area and limited population density at the baseline in 1975. Regions who were showing both a low value and a strong decline of FAP, like Northern Africa (-65.3%) and Western Africa (-74.0%), suggesting the rapid erosion of an already limited forest capital, a pattern confirmed by their significantly negative Sen's slopes (SI). Overall, the data highlighted that population growth was the primary force driving changes in forest area per capita (Fig. S2). The spatial-temporal analysis over several decades showed a notable shift in regions with high forest area per person, with some areas experiencing increases (primarily in Europe) and others showing substantial declines, reflecting the complex interplay between demographic trends and forest change. The phenomenon of decreasing FAP in most regions of the world, except parts of Europe, may be connected to historical, ecological, and socio-economic factors. In Europe, the relatively stable or declining population contrasts with the positive population growth rates in other parts of the world (Fig. S4), where ongoing deforestation in regions like Africa, South America, and Southeast Asia is driven also by global demands for agricultural commodity products[30].

Temporal patterns of Forest Proximate People

As defined in the literature, the metric of Forest Proximate People (FPP) measures the number of people living within or close to a forest[9]. In our study, we assessed the temporal dynamics of FPP (defined as people living within 5 km of a forest) from 1975 to 2020 (Fig. 3).

The selection of 5 km as the threshold distance for defining forest proximity follows recent literature conventions[31], but we acknowledge this is a subjective parameter that may vary depending on research context and regional characteristics. Different distances might be more appropriate depending on factors such as transportation infrastructure, terrain, land use patterns, and cultural practices that influence human-forest interactions. For instance, walking distances in mountainous regions versus plains, or distances considered accessible in densely versus sparsely populated areas may differ substantially. Our computational framework, available in the public repository accompanying this paper, allows for flexibility in calculating FPP using different distance thresholds. This adaptability enables researchers to conduct sensitivity analyses or tailor the proximity definition to specific regional contexts while maintaining the consistent spatial analytical approach used in this study (see Section S6 in SI).

Globally there was an average consistent increase in FPP over the study period, resulting in more people living near forests in 2020 than there were in 1975. This pattern was mainly driven by the overall rise in global population rather than by the expansion of forested areas or by people relocating closer to forests (Fig. S8 in SIs).

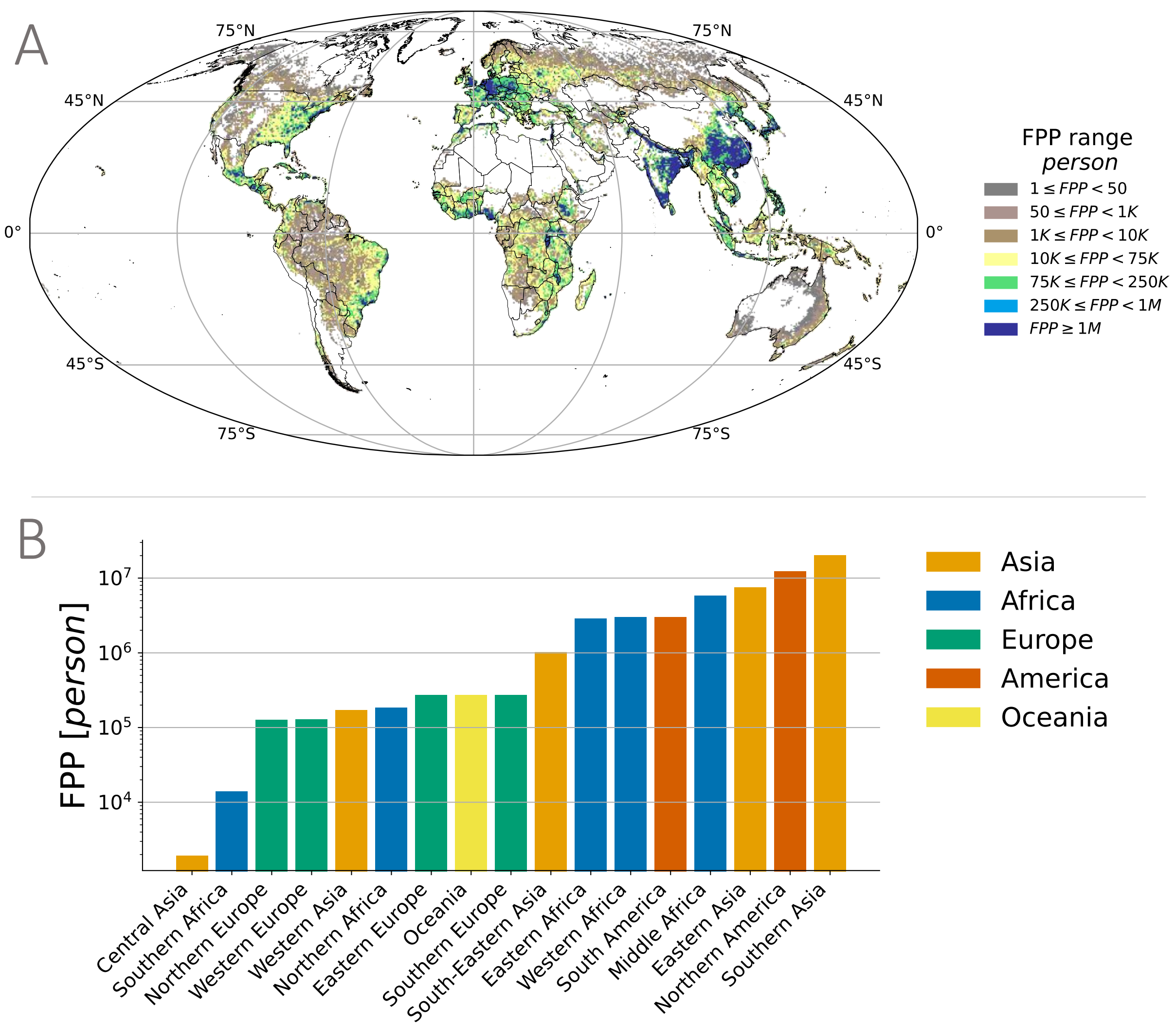

Figure 3. Global distribution of forest proximate people (FPP) in 2020. A) FFP values in 2020 globally and -B) for macroregions.

To factor out the effect of population growth from the trend analysis we defined an index of relative change of the normalized value of FPP (named $\overline{FPP}$), as defined in the Methods section.

Despite the general growth in population numbers, the relative change in the percentage of the total population living near forests showed different patterns across macroregions (Fig. 4A and Fig. 4B). The trend in normalized FPP value highlights that some regions, including Europe, showed positive relative changes even though the absolute FPP values declined (Fig. 3A, Fig. 4A, and Fig. 4B; Fig. S9B).

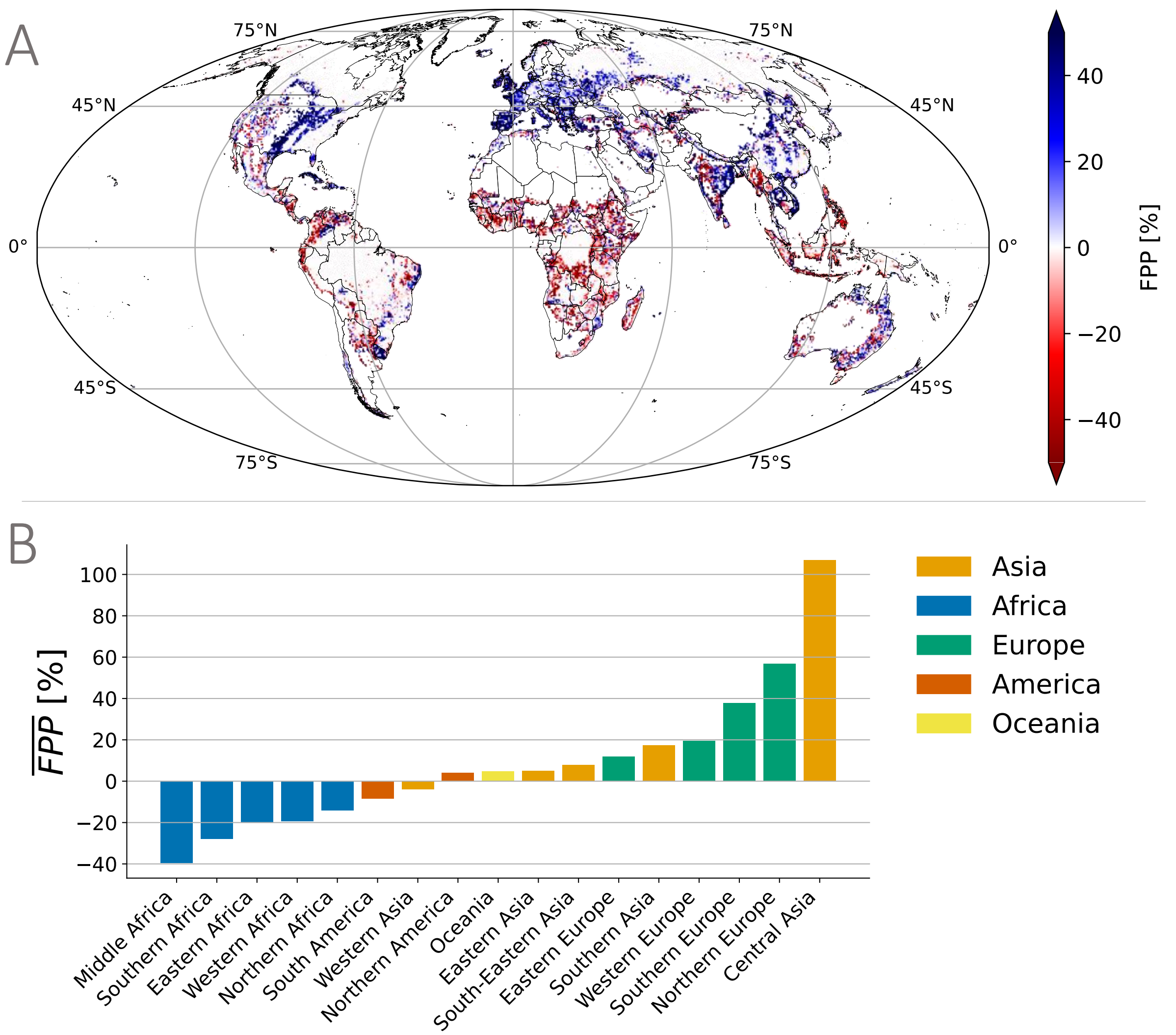

Figure 4. Global change of the relative value of Forest Proximate People ($\overline{FPP}$) between 1975 and 2020. A) Relative change of $\overline{FPP}$ trend between 2020 and 1975 globally. B) Relative change of median values of $\overline{FPP}$ across macroregions between 2020 and 1975.

In contrast, regions showing decline in $\overline{FPP}$, such as African countries, exhibited significant increases in the absolute value of FPP due to population growth (Fig. S6). This suggests that, despite global trends of urbanization and deforestation, the proportion of people living near forests increased between 1975 and 2020. A comparative analysis between Europe and Africa underscores a stark contrast: Europe’s 40% increase in the fraction of the population living near forests may reflect urban expansion into forested areas and a growing appreciation for accessible green spaces, driven by urban planning that integrates green belts and forest reserves. Conversely, Africa’s 40% decrease highlights extensive deforestation and land-use change for agriculture and urban development, which has outpaced efforts to conserve and expand forested areas (Fig. S7D in SI).

We also examined trends in the proportion of FPP across regions using the Sen's slope analysis (detailed in SI). Our analyses reveal an intriguing geographical dichotomy, with results showing strong correlation with the relative change analysis showed above (Fig. S10). African regions and South America show decreasing proportions of their populations living near forests (negative Sen's slopes), alongside declining forest area per person. In contrast, European regions exhibit increasing proportions of forest-proximate populations (positive Sen's slopes) while maintaining stable or increasing forest area per person. This pattern suggests different trajectories of human-forest relationships: in developing regions, diminishing proximity to forest resources may indicate both continued deforestation and changing settlement patterns, while in Europe, the growing integration of forests and human settlements reflects different socioeconomic and land use dynamics. The temporal patterns of $\overline{FPP}$ does not show significant turning points as shown in Figure S11 in SI except for Northern Europe in 1990 and Central Asia in 1995 when a significant increment of $\overline{FPP}$ have been observed.

Temporal patterns of Forest Human Nexus

We introduce the Forest Human Nexus (FHN) index as a novel metric to capture the complex spatial relationship between human populations and forest ecosystems (Fig. 5). The Forest Human Nexus (FHN) provides an integrated spatial measure that combines the number of people living near forests, the forest area available, and the proximity (i.e., distance) of human populations to the nearest forested areas (see Methods section and SI). The FHN value ranges from 0 to 1, where 1 represents maximum human-forest proximity (i.e., a single person residing in a window where all surrounding pixels are forested), and 0 indicates the absence of both forests and people. Over the 45-year period (1975–2020), FHN exhibited significant spatial and temporal variations across different macroregions (Fig. 6A).

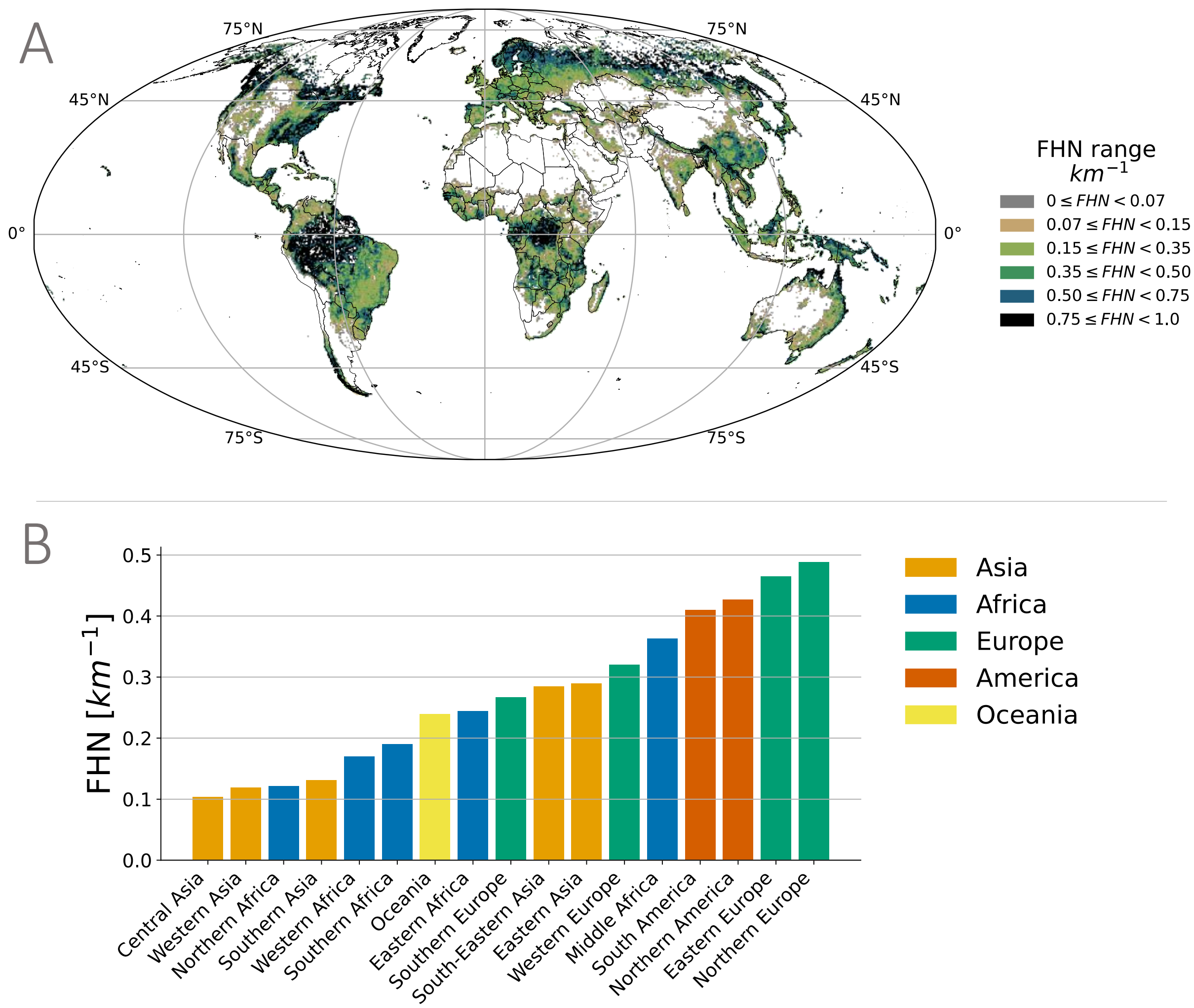

Figure 5. Global distribution of forest human nexus (FHN) in 2020. A) Global and B) macroregions distributions of FHN in 2020, highlighting areas with varying degrees of human proximity to forests. Regions with high FHN values (e.g., Eastern Europe and North America) indicate densely populated areas with substantial forest cover nearby. In contrast, low FHN values reflect regions where human settlements are relatively distant from forested areas, revealing a heterogeneous global pattern of human-forest proximity

Positive trends in FHN over time suggest increasing human proximity to forests, which may be influenced by forest regrowth, urban planning that integrates green spaces, or demographic shifts toward forested regions. Conversely, negative FHN trends—observed primarily in parts of Africa and South America—suggest an increasing spatial separation between human populations and forests. Multiple factors could underlie these shifts, including forest loss, deforestation-related land-use change, and population dynamics affecting settlement patterns (e.g., rural to urban migration).

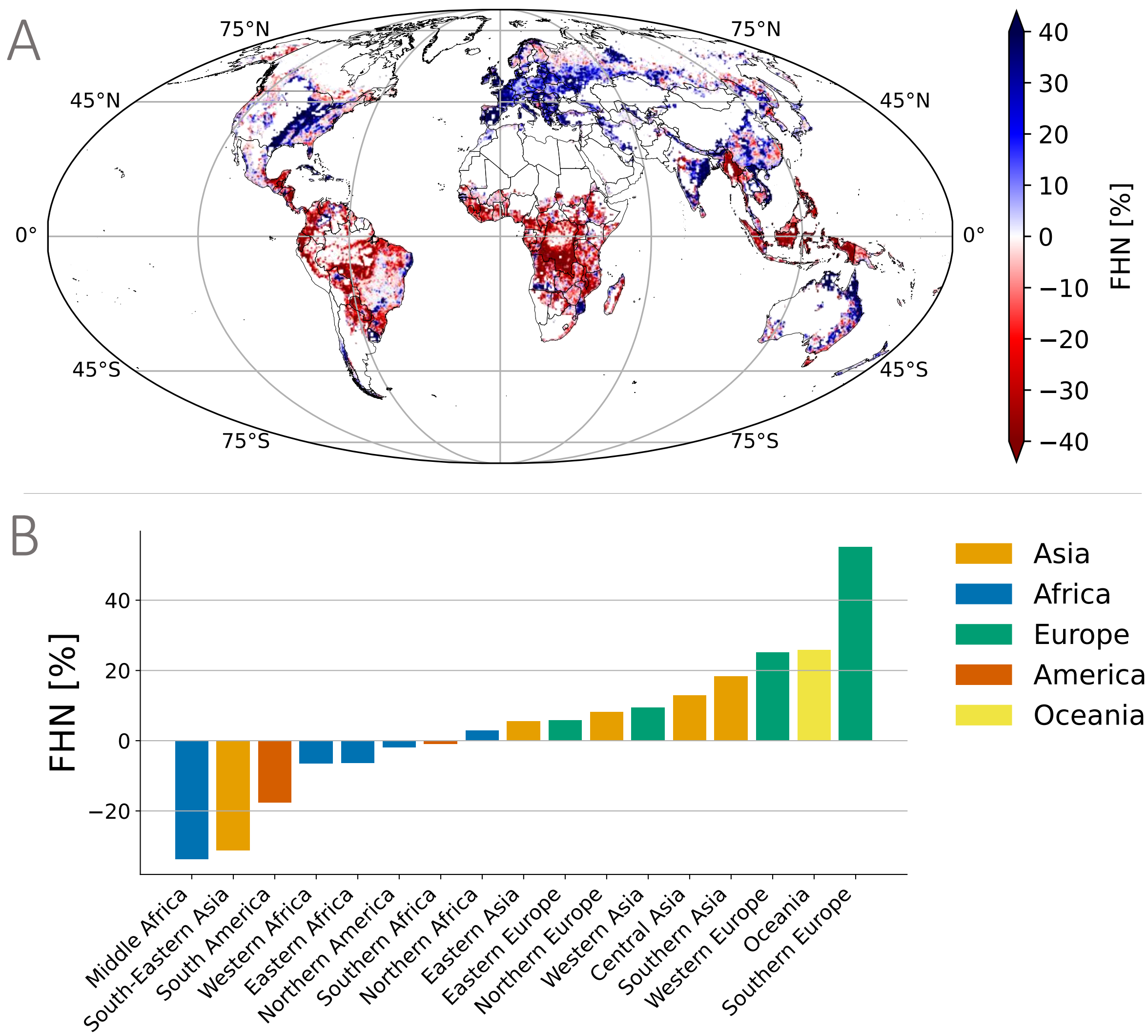

Figure 6. Historical distribution of the forest human nexus (FHN) indicator from 1975 to 2020. A-B) Relative change of the median values of FHN globally and across microregions. Positive FHN trends indicate that human populations have moved closer to forests.

At the regional scale, contrasting trajectories emerged: South-Eastern Asia (-31.2%), Middle Africa (-33.7%), and South America (-17.5%) (in red, Fig. S17) exhibited a consistent decline in FHN, reflecting increasing spatial distances between human populations and forests. In contrast, Southern Europe (+55.2%), Western Europe (+25.1%), and Southern Asia (+18.3%) (in blue, Fig. S17) showed substantial increases in FHN values, suggesting that human-forest proximity was significantly increased in these regions. Our Sen's slope analysis (detailed in SI) strongly corroborated these relative change findings, confirming the robustness of these regional patterns. These patterns were observed consistently across major global forest biomes, including the Amazon region, Congo Basin, boreal forests, and Southeast Asian tropical forests, highlighting the value of our consistent 50km × 50km resolution approach for comparative global analysis.

Further analyses of FHN distributional shifts (Fig. S18 and Fig, S19 in SI) demonstrated strong correlation with the FHN trends (Fig. S20 in SI) and relative changes (Fig. 6). This supplementary analysis confirmed our primary finding that different regions experienced divergent trajectories in human-forest relationships. Rather than focusing on specific regional comparisons, these patterns collectively emphasized the need for regionally tailored conservation and sustainable management approaches that account for the distinct socio-ecological contexts of each region. The observed spatial variations in FHN highlight potential implications for ecosystem service flows, biodiversity conservation, and climate change mitigation efforts across diverse landscapes worldwide.

For most of the regions analyzed, the FHN trends followed relatively consistent trajectories without dramatic turning points (see Fig. S21 in SI), suggesting gradual, long-term shifts in human-forest relationships driven by sustained socioeconomic and environmental forces. South America exhibited a steady decline without significant inflection points, indicating a continuous degradation of forest-human relationships over the 45-year period. Similarly, Western Europe, Western Africa, and Eastern Asia displayed consistent directional trends with minimal deviation.

However, we observed notable turning points in several key regions. Southern Asia presented one of the most dynamic patterns, with multiple turning points (1985, 1990, 1995, 2005, 2015) creating a distinctive wave-like trajectory that ultimately results in a positive trend. Middle Africa showed a critical inflection point around 1995-2000 where the previously moderate decline accelerates dramatically, suggesting a fundamental shift in forest-human dynamics during this period. Southern Europe displayed particularly significant turning points, with a significant upward trajectory beginning around 1985-1990 that transformed the region from having modest FHN values to exhibiting the strongest positive trend among all regions.

Central Asia stood out with a clear reversal around 1985, transitioning from a negative to a strongly positive trend. In Northern Africa, the turning point around 2010 signaled a shift from decline to recovery, suggesting recent changes in policy or land use patterns. Oceania exhibited turning points around 2005 and 2010 within its consistently improving trajectory.

These regional variations in turning points highlight the complex, non-linear nature of human-forest relationships and suggest that while some regions experience gradual, predictable changes, others underwent more dramatic shifts that may reflect specific policy interventions, socioeconomic transformations, or environmental threshold effects. The identification of these turning points provides valuable insights for understanding historical dynamics and potentially forecasting future trajectories of forest-human relationships across different global contexts.

Discussion

Forests, covering 31% of the Earth's surface, are critical to maintaining environmental stability, biodiversity, and providing essential ecosystem services[32]. While previous studies have explored various aspects of human-forest interactions[9,33], the three indicators presented in this study – Forest Area per Person (FAP), Forest Proximate People (FPP), and Forest Human Nexus (FHN) – offer complementary perspectives on human-forest relationships that together provide a more comprehensive understanding than any single metric alone. While FAP quantifies the theoretical forest resource availability per capita, FPP measures the absolute number of people with potential physical access to forests, and FHN integrates spatial proximity with population distribution. These indicators exhibit different regional applicability and significance. In densely populated regions with limited forest cover (e.g., parts of South Asia), FHN proves particularly valuable for identifying areas where even small forest fragments maintain important connections to human settlements. In contrast, in regions with extensive forest cover but low population density (e.g., boreal regions), FAP may better represent resource availability. The regional differences in indicator performance reflect the diverse socio-ecological contexts of human-forest interactions worldwide[34]. For instance, in Europe, increasing FHN despite stable FAP reveals improved spatial integration of forests and human settlements through targeted landscape planning, while in parts of Southeast Asia, declining FHN despite high absolute FPP numbers signals the spatial disconnection occurring despite large populations still living near forests[35]. By analyzing these indicators in conjunction, policymakers can develop more nuanced, region-specific approaches to sustainable landscape management that account for both the quantitative aspects of forest resources and their spatial relationship to human populations[36].

In particular, the Forest Human Nexus (FHN) indicator is as a geospatial metric that quantifies human proximity to forests over a 45-year period (1975--2020) at a global scale. The FHN indicator integrates multiple dimensions of spatial human-forest relationships, including the number of people living near forests, forest area per capita, and the average distance from human populations to the nearest forest. This last metric, calculated as the inverse of the weighted average distance from human settlements to forests, provides a spatially explicit measure of how human-forest proximity has evolved over time. Our analysis reveals significant regional variations in FHN, reflecting heterogeneous spatial patterns of human-forest proximity across the globe. The global distribution of FHN in 2020 (Fig. 3A) demonstrates substantial geospatial heterogeneity, with some regions maintaining close human-forest proximity, while others exhibited increasing spatial separation. These patterns align with findings from previous studies on global forest fragmentation[37] and urbanization trends[38], which have documented increasing separation between human settlements and intact forest ecosystems in many developing regions. Temporal trends in FHN (Fig. 3B) indicate notable shifts in human-forest proximity over the past four decades. Regions such as Southern Europe and Western Europe experienced positive trends, suggesting that human populations in these areas have moved closer to forests. These trends are consistent with documented European forest transition dynamics[39], including reforestation efforts, urban planning strategies that incorporate green spaces, and demographic shifts toward rural areas with greater forest cover[40]. Conversely, negative trends were prevalent in Southeast Asia, Middle Africa, and South America, indicating an increasing spatial separation between human populations and forests. This separation aligns with well-documented land-use changes in these regions, including large-scale deforestation for agricultural expansion[41], urbanization in non-forested areas[42], and population growth concentrated in urban centers distant from remaining forest fragments[43]. The distributional analysis of FHN changes (Fig. 4) provides additional insight into regional disparities in human-forest proximity. In Southern Europe, FHN distributions shifted rightward, indicating that a greater proportion of the population lived closer to forests over time, consistent with observed forest expansion in abandoned rural areas[44] and increasing preferences for forest-proximate living. In contrast, Southeast Asia exhibited a leftward shift, signifying an increasing spatial distance between human populations and forests.

This pattern is consistent with research documenting rapid urbanization[45], forest conversion to plantation agriculture[46], and the concentration of population growth in coastal and peri-urban areas rather than forested regions[47].

While the FHN metric provides valuable spatial insights into human-forest relationships, it is important to acknowledge several limitations. First, our analysis relies on remote sensing-based forest definitions that may not fully capture forest quality, degradation, or ecological functionality. Second, the FHN does not directly measure actual forest resource use or dependency by adjacent populations, which can vary significantly based on socioeconomic, cultural, and institutional factors. Looking beyond these limitations, the FHN framework has significant potential for broader applications beyond spatial proximity analysis. The metric can be integrated into more comprehensive assessments of bidirectional forest-human interactions by combining it with socioeconomic data, ecosystem service valuations, and governance indicators. Such integration would enable the FHN to contribute meaningfully to monitoring progress toward multiple Sustainable Development Goals (SDGs), particularly those related to poverty elimination (SDG 1), food security (SDG 2), health and wellbeing (SDG 3), clean water (SDG 6), climate action (SDG 13), and terrestrial ecosystems (SDG 15). For example, by overlaying FHN with poverty data, researchers could identify critical areas where forest conservation might impact vulnerable populations, enabling more equitable policy design. Similarly, combining FHN with biodiversity hotspot information could highlight priority zones for conservation that balance human needs with ecological imperatives. Future research should develop these integrative approaches to transform the FHN from a spatial proximity measure into a comprehensive tool for understanding and managing the complex, bidirectional interactions between forests and human societies in pursuit of sustainable development.

These findings emphasize the importance of spatially explicit assessments of human-forest proximity in understanding long-term land-use changes and their potential implications[48]. The observed global heterogeneity in FHN trends highlights the complex interplay between socioeconomic development, environmental policies, and demographic shifts in shaping human-forest relationships[49]. Regions experiencing declining FHN values may face greater challenges in maintaining ecosystem service flows to human populations[50], while also potentially reducing anthropogenic pressures on remaining forest ecosystems[51]. As we look to the future, the FHN can serve as a geospatial tool for researchers, policymakers, and conservationists in monitoring and assessing spatial trends in human-forest proximity. Future research should explore the socioeconomic and environmental drivers underlying the observed FHN patterns, as well as the implications of changing human-forest proximity for biodiversity conservation[52], ecosystem service provision[53], and human well-being[54]. Ensuring a sustainable balance between human presence and forest ecosystems will be crucial for maintaining environmental integrity and long-term landscape resilience in a rapidly changing world.

Methods

This study combines population and land use data. We took the population layer from the EU Global Human Settlement Layer (GHSL)[29], while for land use data we made use of the HILDA+ dataset[29]. We used the GUIDOS toolbox[55] to compute the distance to forest layer. In this study, we analyzed the relationship between forest land use and population distribution starting from a spatial resolution of 1 km.

To achieve this, we employed a moving window approach with a window size of 50 km by 50 km. This method allowed us to compute spatially explicit metrics that captured the interaction between forest area and population across the landscape. This moving window technique is commonly used in landscape ecology and spatial analysis to capture multi-scale patterns and processes across heterogeneous landscapes[56]. We selected the 50 km × 50 km window size as an optimal compromise between fine-scale detail and computational efficiency, aligning with similar spatial analyses in global land-use studies. Within each window, we computed metrics using a neighborhood function that considers all pixels within the defined window, allowing us to capture spatial relationships beyond administrative boundaries. This approach is particularly suitable for analyzing human-forest interactions as it accounts for the spatial influence of forests that extends beyond their immediate boundaries. The moving window method allows us to quantify spatial patterns at a consistent scale across diverse landscapes, facilitating global comparisons while maintaining sensitivity to regional variations in human-forest distributions.

Specifically, we calculated three key metrics within each window: Forest Area per Person (FAP), Forest Proximate People (FPP), and the new Forest Human Nexus (FHN) indicator. The FAP metric is the sum of the total forest area within the window divided by the total population of the window's constituent pixels. The number of FPP is the number of people living within a given distance to forest inside each window. For the FHN metric, we assess the inverse of the average distance to the nearest forest area, weighted by population density, by computing the sum of the product of population and distance for each pixel and dividing this by the total population within the window. Additionally, we aggregated the data obtained from these computations at continental, regional, and country levels, enabling broader-scale analyses and comparisons that could inform policy and management decisions at multiple geographic scales.

Population dataset

The Global Human Settlement Layer provides the population dataset with a resolution of 1km where residential population estimates between 1975 and 2020 in 5-year intervals and projections to 2025 and 2030. The dataset is available online (https://data.jrc.ec.europa.eu/collection/ghsl).

Land use dataset

For the land use dataset, we used the HILDA+ (HIstoric Land Dynamics Assessment+) global dataset on annual land use/cover change between 1960 and 2019 at 1 km spatial resolution. It is based on a data-driven reconstruction approach and integrates multiple open data streams (from high-resolution remote sensing, long-term land use reconstructions and statistics). It covers six generic land use/cover categories: 1: Urban areas, 2: Cropland, 3: Pasture/rangeland, 4: Forest, 5: Unmanaged grass/shrubland, 6: Sparse/no vegetation. The dataset is available online (https://ceos.org/gst/HILDAplus.html).

Forest definition

HILDA defines forests as land spanning more than 0.5 hectares with trees higher than 5 meters and a canopy cover of more than 10%, or trees able to reach these thresholds in situ. This definition aligns with the Food and Agriculture Organization (FAO) land use guidelines and incorporates both natural forests and plantations. Forests in HILDA+ are mapped using various high-resolution datasets, including the Copernicus LC100 Global Land Cover map, which provides detailed and accurate spatial representation of forest cover as of the base year 2015.

The forest category in HILDA+ includes the following characteristics:

- Tree Density and Height: Areas dominated by trees that meet the height and canopy cover thresholds.
- Land Use: Both primary forests (undisturbed by human activity) and secondary forests (areas regenerating from previous deforestation) are included.
- Spatial Coverage: The dataset captures forests at a 1 km x 1 km grid resolution, ensuring detailed spatial representation and allowing for the analysis of forest dynamics over time.

The forest classification in HILDA+ does not explicitly differentiate between various types of green areas with trees (such as urban parks or agroforestry systems). Instead, it focuses on areas where trees form a significant component of the land cover, contributing to the understanding of broader forest dynamics and their role in global land use changes. By utilizing HILDA+ for our analysis, we ensured a consistent and harmonized approach to defining and mapping forests, enabling a robust examination of forest cover changes and their implications for environmental and climatic processes.

Limitations: While the HILDA+ dataset provided a robust framework for our analysis, several limitations should be considered. First, the integration of multiple datasets with varying spatial and temporal resolutions introduces uncertainties. The harmonization process may sacrifice finer details and create discrepancies in land use/cover classification. Second, HILDA+ relies on a static base year (2015) for initial classification, which may not accurately reflect recent forest dynamics. Changes are modeled based on historical trends that may not capture emerging land use patterns. Additionally, remote sensing data inherently contains errors due to atmospheric conditions, sensor limitations, and misclassification issues that affect forest mapping accuracy. Forest definition variability represents another significant source of uncertainty. The HILDA+ dataset follows FAO forest definitions based on tree height, canopy cover, and minimum area thresholds, but alternative definitions emphasizing different structural characteristics, ecological functions, or land-use practices exist[57]. These definitional differences can significantly impact forest area estimates and consequently affect FPP numbers and spatial patterns. For example, definitions with higher canopy cover thresholds would identify fewer forested areas, potentially underestimating FPP values, while more inclusive definitions might overestimate forest proximity relationships. Furthermore, HILDA+ does not differentiate between natural forests and plantations, which have distinct ecological characteristics and human-forest interaction patterns. These various sources of uncertainty should be carefully considered when interpreting our results and when comparing our FPP estimates with those from other studies and datasets.

Forest area per person

We defined the Forest Area per Person (FAP) raster by summing, in each window, the forest pixels and dividing by the sum of the population living in each pixel of the original datasets.

$$FAP_w = \frac{\sum_j F_j}{\sum_j P_j}$$

Where *P* and *F* are respectively the population layer and the forest data at 1km resolution. For each window at 50km resolution, FAP_w is defined by the above equation where P_j and F_j are respectively the amount of people living the in the pixel *j* and the forest pixel (0 non-forest, 1 forest) in the pixel *j*. The unit of measure is $km^2 \cdot person^{-1}$. This also includes the people that are not living in a forest.

Distance to forest

We used GUIDOS toolbox[55] to generate the distance to forest dataset, which we then used to compute the subsequent metrics. Starting from the land use dataset, we set each pixel 0 as background, 1 if the land use was a non-forest pixel, and 2 if it was a forest pixel. On any location of this map, the pixel value displays the shortest Euclidean distance to the nearest forest boundary. To assess the distance to forest we set as zeros all the values inside the forests including the forest boundary and as positive values all the distances in non-forest areas. The unit of measurement is km.

Forest proximate people

The Forest Proximate People (FPP) indicator counts the total population living within a particular distance from a forest. By considering the population dataset and distance data, we were able to assess the number of FPP at different distances in a moving window of 50km x 50km. As for the other indicators, we aggregated the 1km resolution data to a 50km x 50km resolution data. In each window, we computed FPP as the number of persons living within a given distance to the forest.

In order to factor out the effect of population growth from the trend analysis we define an index of relative change of the normalized value of FPP defined as:.

$$\overline{FPP}[\%] = 100 * (\overline{FPP}(2020) - \overline{FPP}(1975))/\overline{FPP}(1975)$$

where $\overline{FPP}(year) = FPP(year)/P(year)$.

Forest human nexus

The Forest Human Nexus (FHN) indicator represents the inverse of the average weighted distance from human. We computed the FHN indicator globally using a moving window approach as described in the Supplementary Information. For each window *w* (50x50), the FHN_w, is defined as

$$FHN_w = \frac{\sum_j P_j}{\sum_j P_j \cdot (D_j + 1)}$$

where *j* is the cell *j* of the window *w*, P_j is the number of people in the cell *j* and D_j is the Euclidean distance from the cell *j* to the closest forest. The unit of measure of FHN is km^{-1} and this value is bounded between 0 and 1, which means that 1 is highest value where only one person lives in an area covered by 100% of forest, while 0 corresponds to a place with no forest and/or where no people live.

Data availability

All the data used in this research is public available: the original datasets and the aggregated data, as well as the Python codes used for the study, are stored in a public available repository (https://github.com/emanuelemassaro/Forest_Human_Nexus).

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Author contributions

E.M. and A.C. designed the research; E.M. performed the research, collected and analyzed the data; All the authors revised and discussed the results of the research, wrote, and revised the paper.

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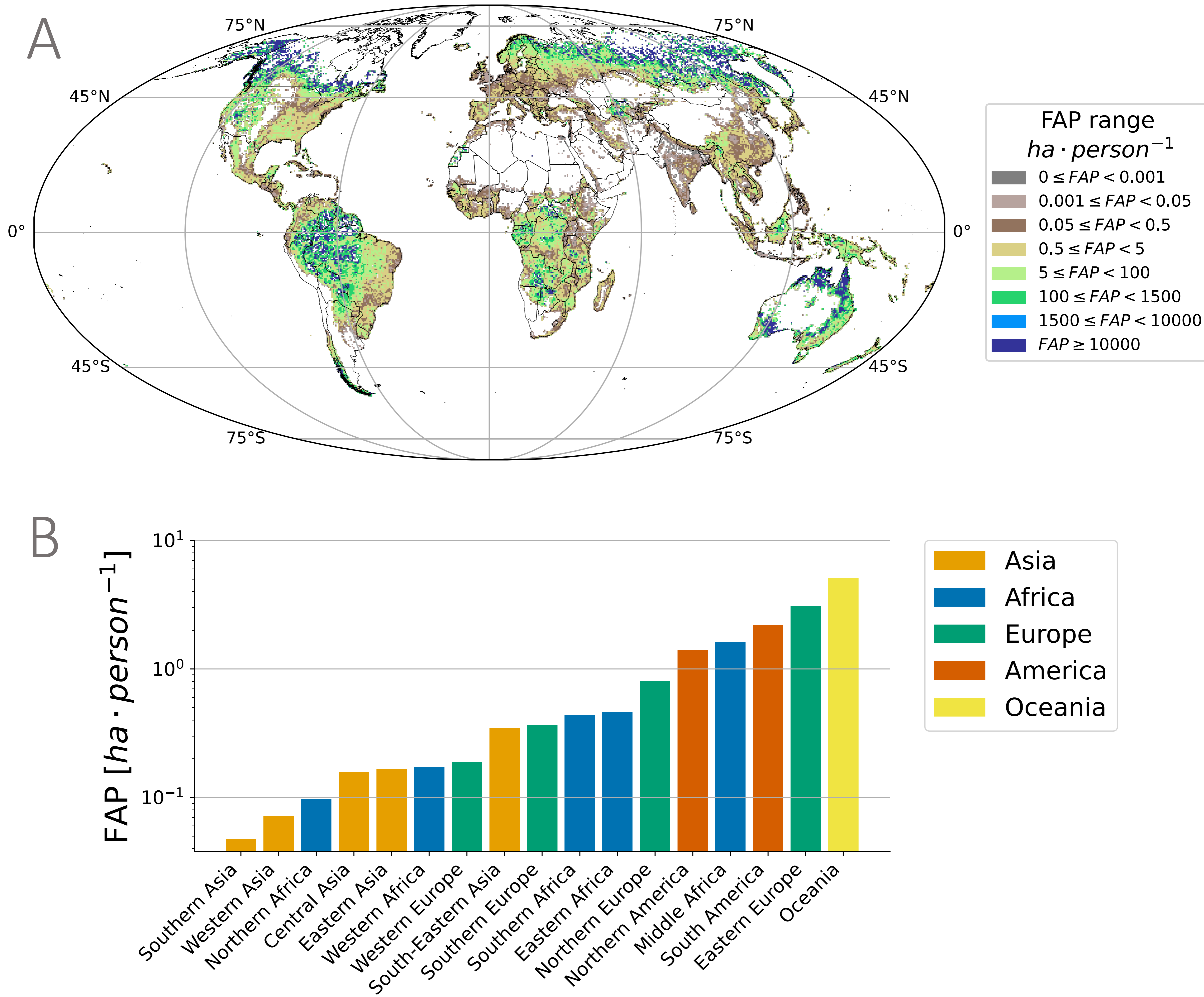
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FAP range
$ha \cdot person^{-1}$
$0 \leq FAP < 0.001$
$0.001 \leq FAP < 0.05$
$0.05 \leq FAP < 0.5$
$0.5 \leq FAP < 5$
$5 \leq FAP < 100$
$100 \leq FAP < 1500$
$1500 \leq FAP < 10000$
$FAP \geq 10000$
B
FAP [$ha \cdot person^{-1}$]
10^{1}
10^{0}
10^{-1}
Asia
Africa
Europe
America
Oceania
Southern Asia
Western Asia
Northern Africa
Central Asia
Eastern Asia
Western Africa
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South-Eastern Asia
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Northern America
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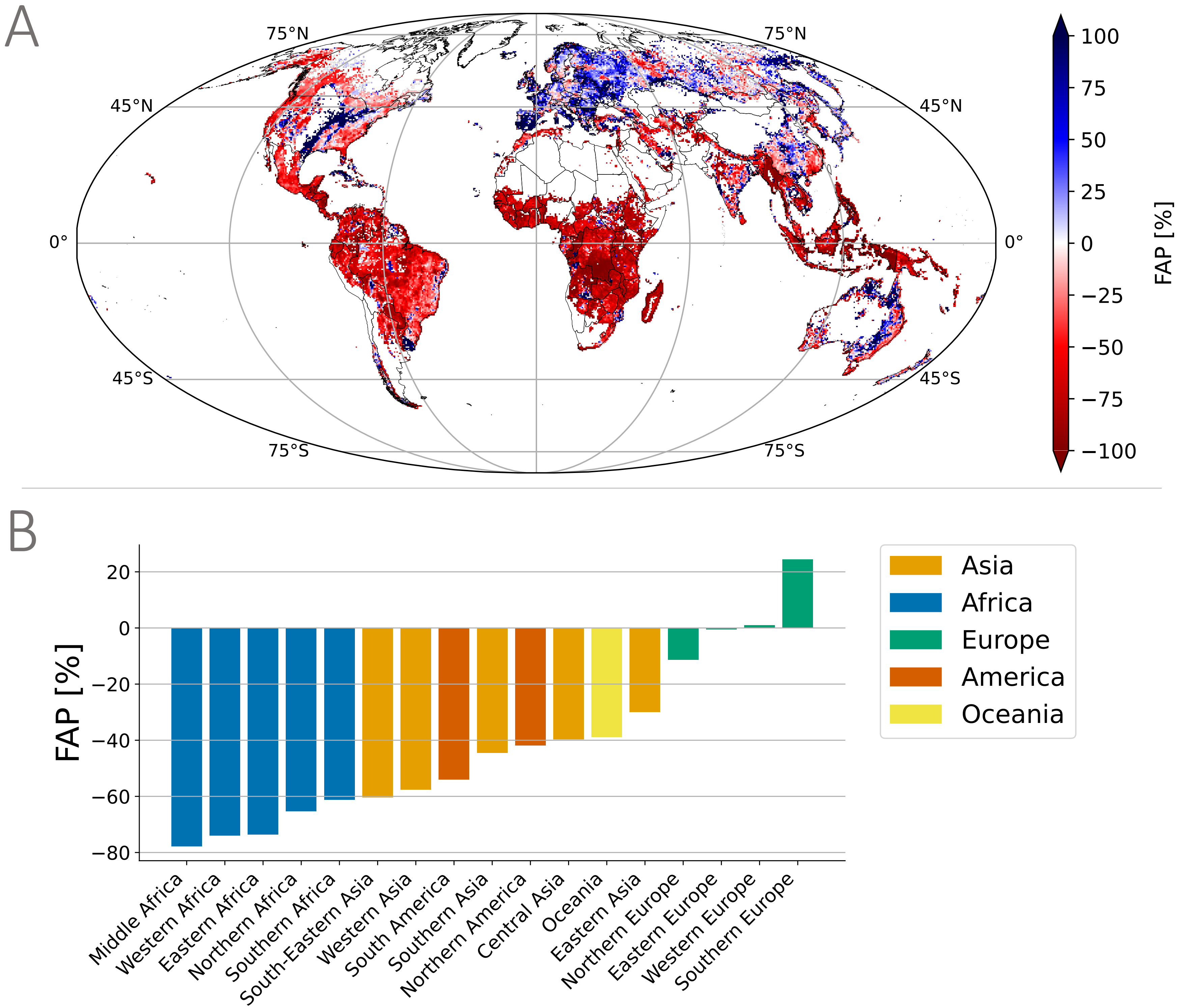

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50
25
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−25
−50
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−100
FAP [%]
B
20
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−40
−60
−80
FAP [%]
Middle Africa
Western Africa
Eastern Africa
Northern Africa
Southern Africa
South-Eastern Asia
Western Asia
South America
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Southern Europe
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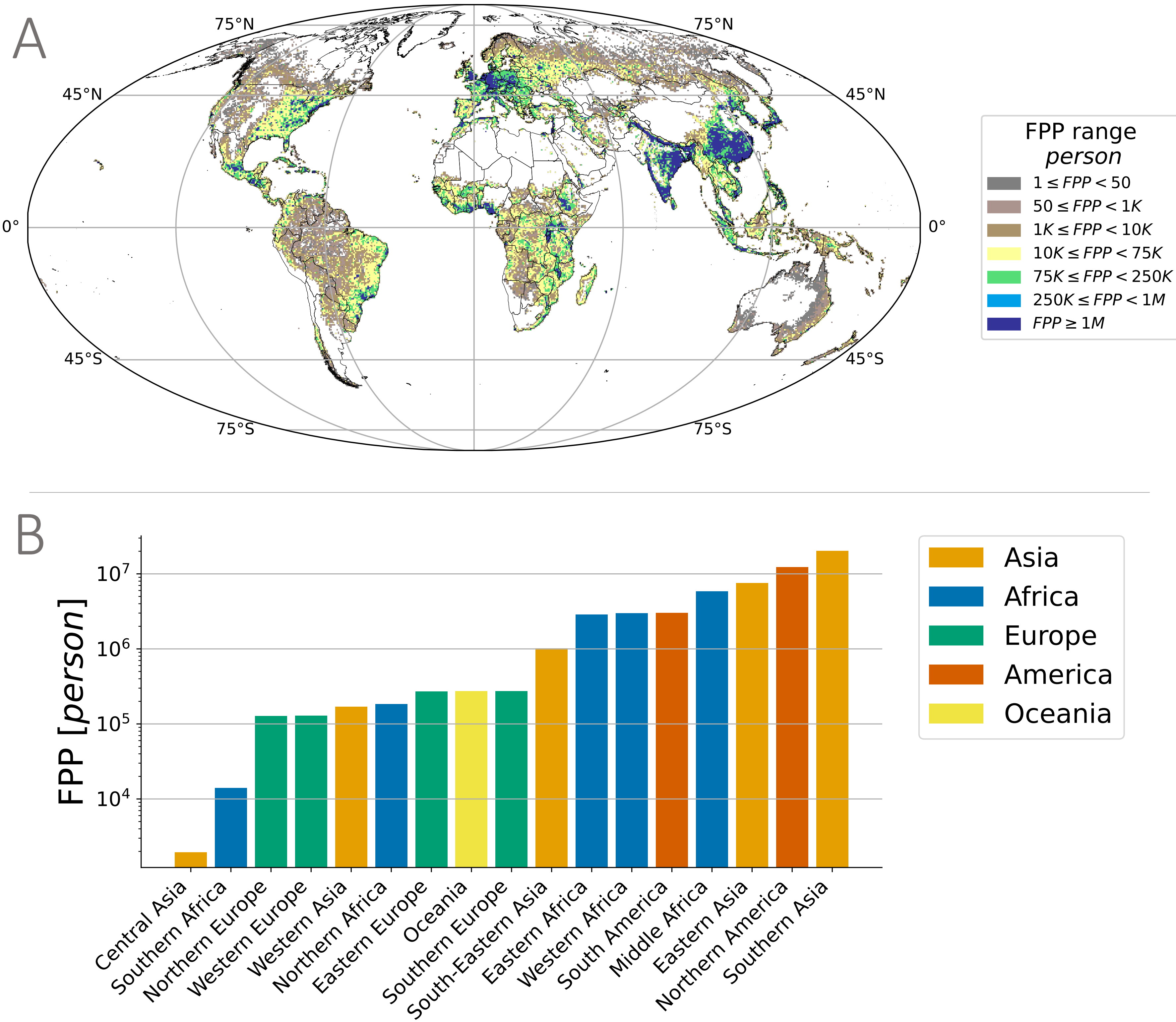
Europe
America
Oceania



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75°S
FPP range
person
1 ≤ FPP < 50
50 ≤ FPP < 1K
1K ≤ FPP < 10K
10K ≤ FPP < 75K
75K ≤ FPP < 250K
250K ≤ FPP < 1M
FPP ≥ 1M
B
FPP [person]
10^7
10^6
10^5
10^4
Asia
Africa
Europe
America
Oceania
Central Asia
Southern Africa
Northern Europe
Western Europe
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Western Africa
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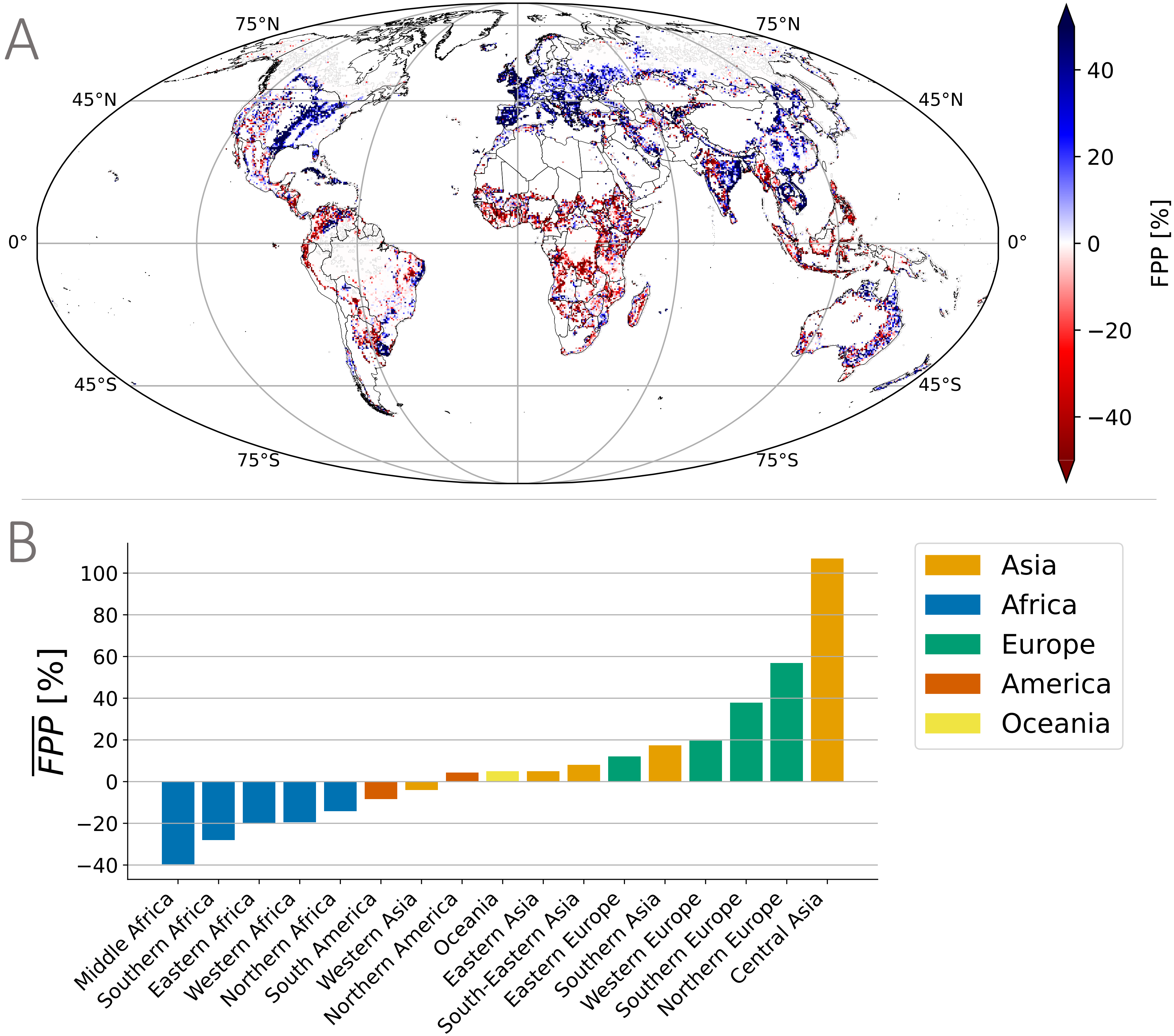

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FPP [%]
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60
40
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0
−20
−40
$\overline{FPP}$ [%]
Asia
Africa
Europe
America
Oceania
Middle Africa
Southern Africa
Eastern Africa
Western Africa
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South-Eastern Asia
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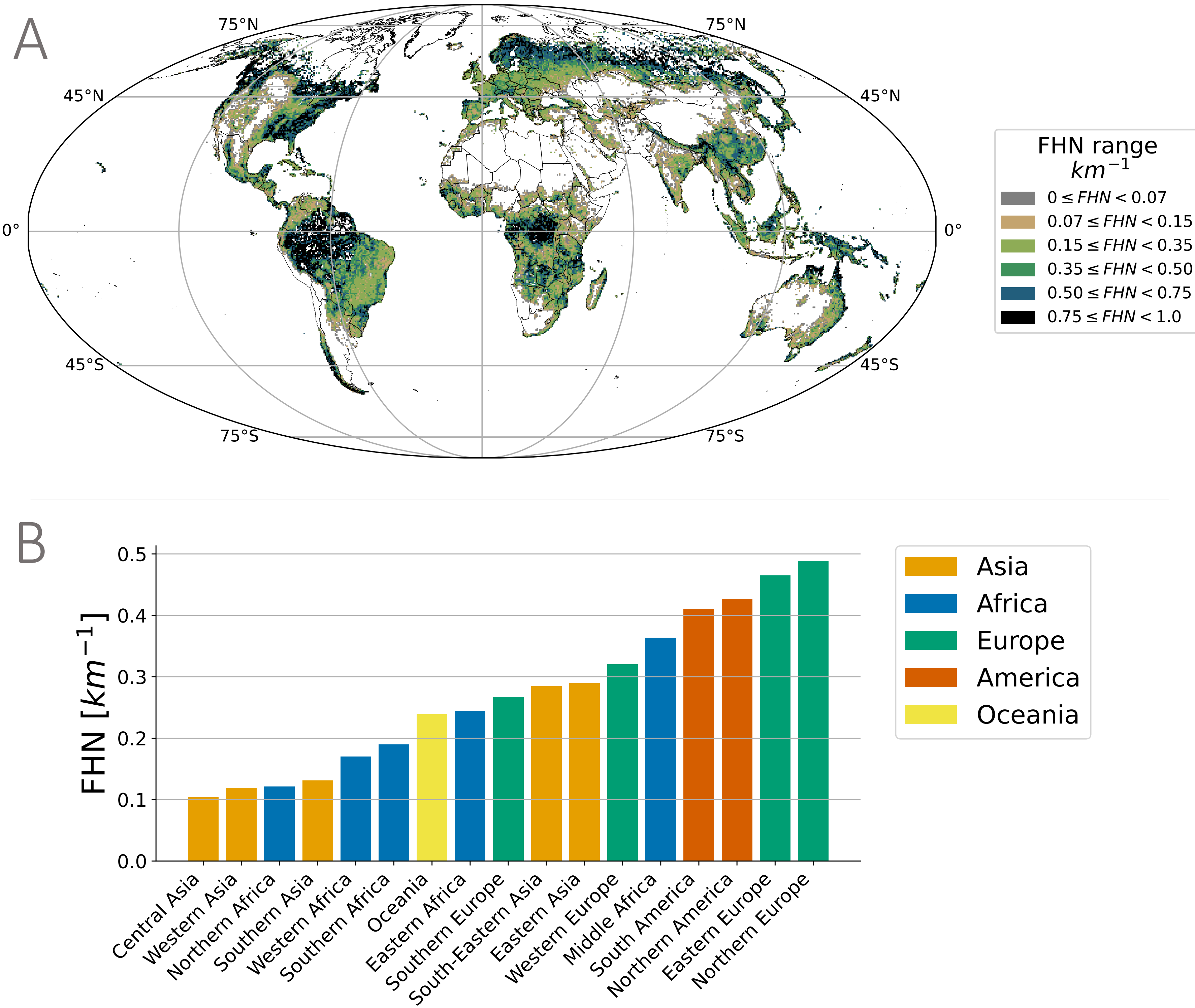
Northern Europe
Central Asia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75°S
FHN range km^{-1}
$0 \leq FHN < 0.07$
$0.07 \leq FHN < 0.15$
$0.15 \leq FHN < 0.35$
$0.35 \leq FHN < 0.50$
$0.50 \leq FHN < 0.75$
$0.75 \leq FHN < 1.0$
B
FHN [km^{-1}]
0.0
0.1
0.2
0.3
0.4
0.5
Central Asia
Western Asia
Northern Africa
Southern Asia
Western Africa
Southern Africa
Oceania
Eastern Africa
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Middle Africa
South America
Northern America
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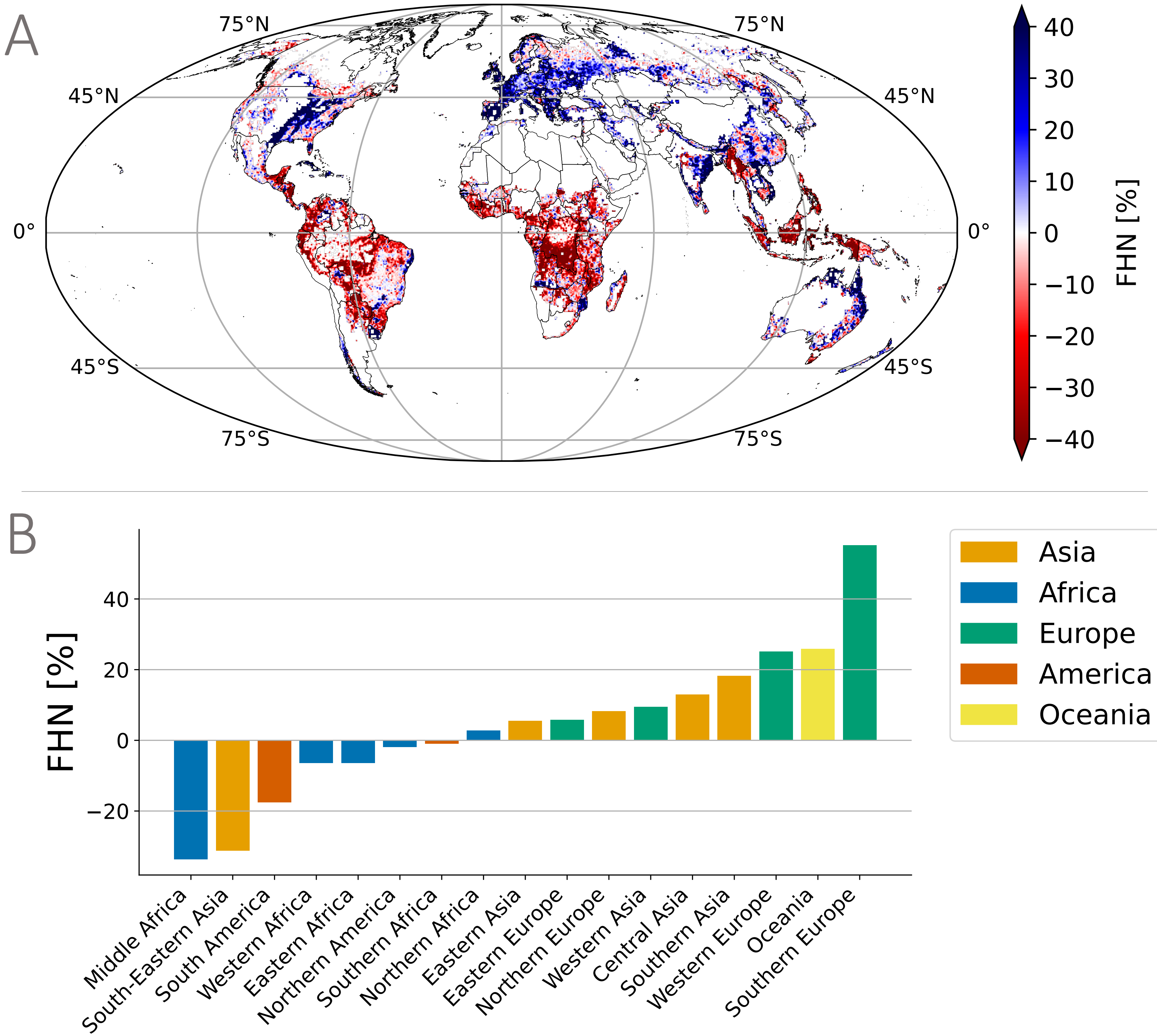
Oceania



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75°S
40
30
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10
0
−10
−20
−30
−40
FHN [%]
B
FHN [%]
40
20
0
−20
Middle Africa
South-Eastern Asia
South America
Western Africa
Eastern Africa
Northern America
Southern Africa
Northern Africa
Eastern Asia
Eastern Europe
Northern Europe
Western Asia
Central Asia
Southern Asia
Western Europe
Oceania
Southern Europe
Asia
Africa
Europe
America
Oceania